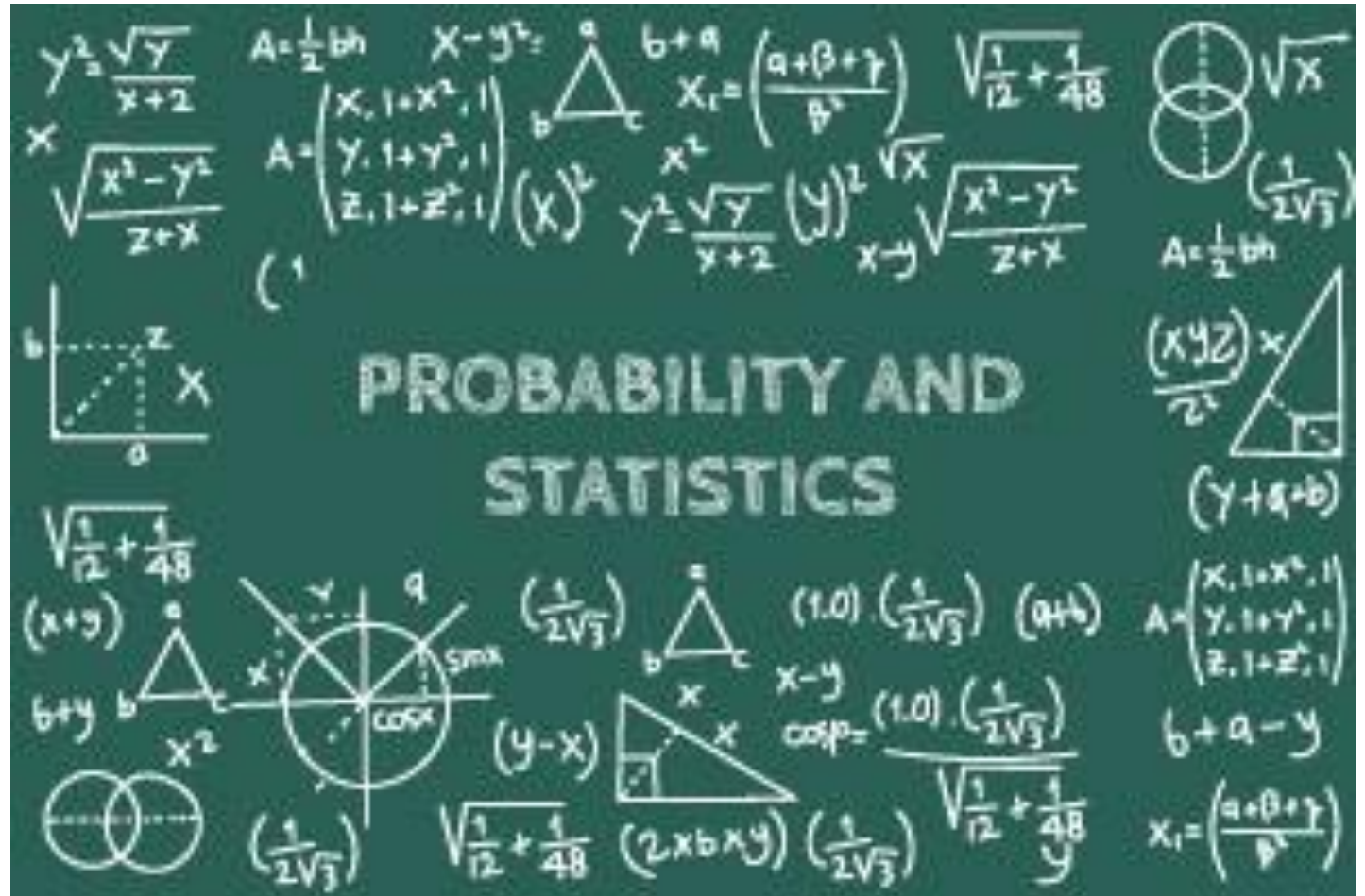


Lec # 09  
ES111

# Total Probab. Bayes' Rule

Book Reading: Sec. 2.7

@M. Tahir Naseem



# Discussion Plan

1. **Total** Probability

2. **Bayes'** Rule

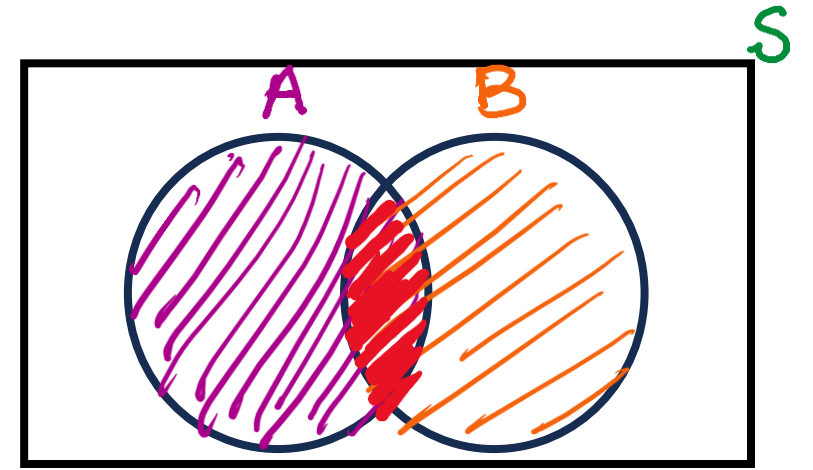
(Sec. 2.7)



## The product rule

$$P(A \cap B) = P(B) P(A|B)$$

$$P(A \cap B) = P(A) P(B|A)$$

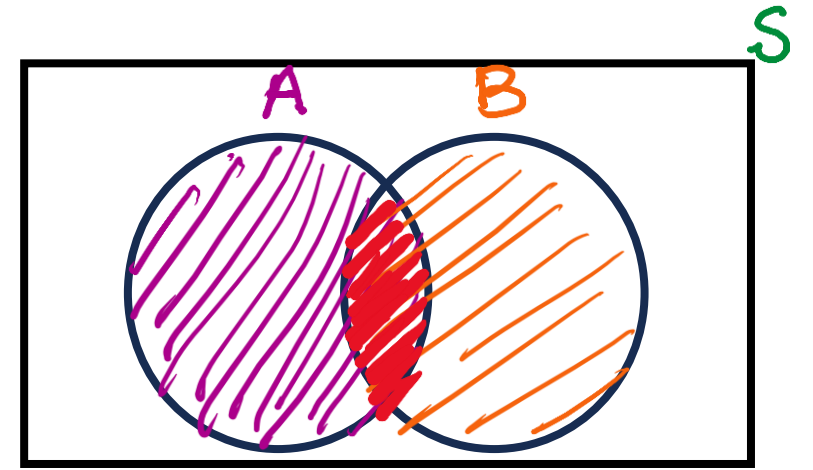


## The product rule

$$P(A \cap B) = P(B) P(A|B)$$

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For Independent Events



## The product rule

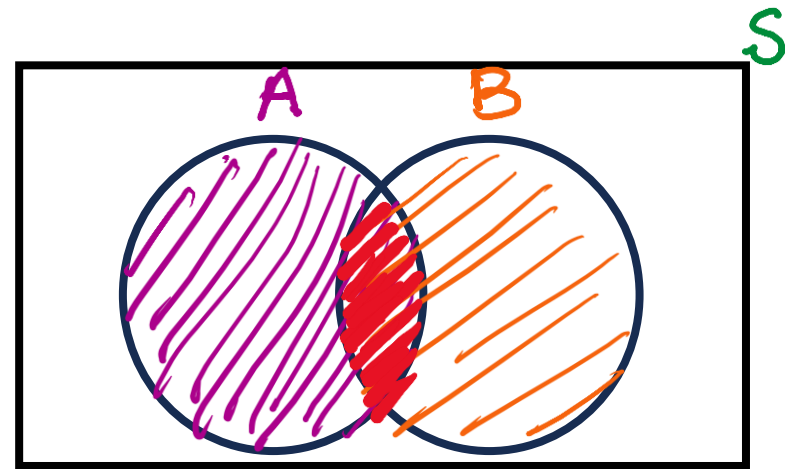
$$P(A \cap B) = P(B) P(A|B)$$

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## For Independent Events

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$



## The product rule

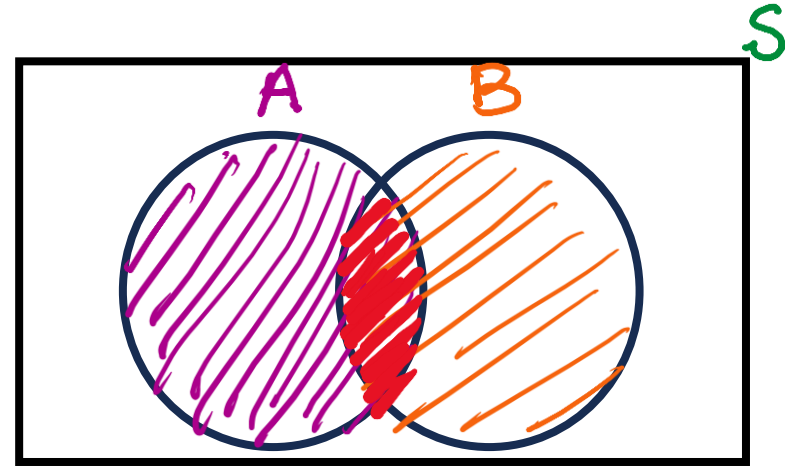
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$$P(A|B) = P(A)$$

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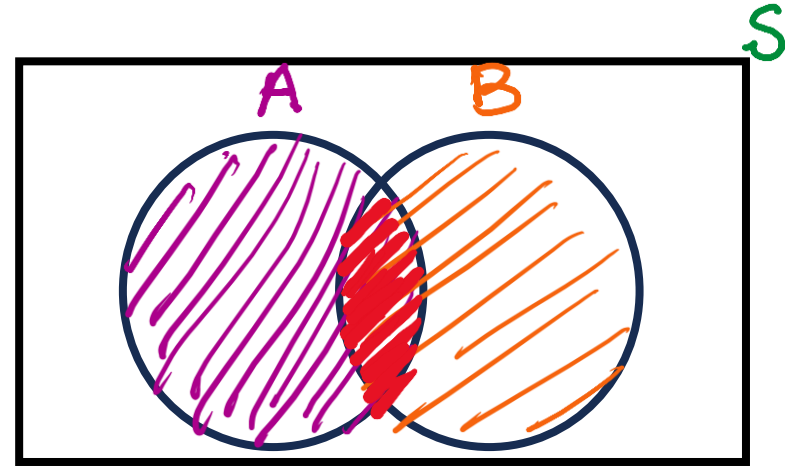
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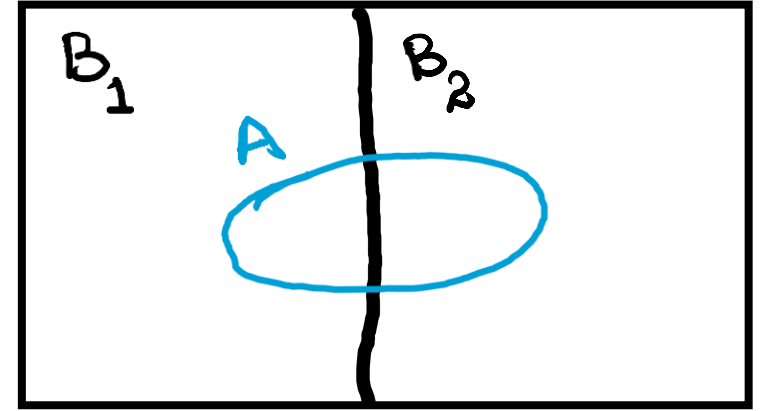


# Total probability Law



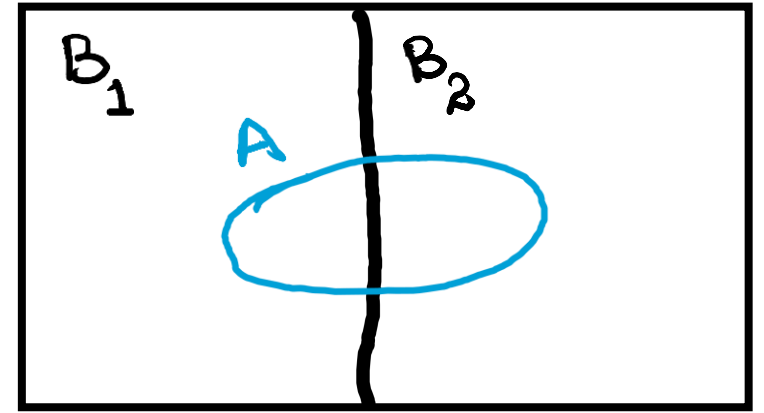
# Total probability Law

S



# Total probability Law

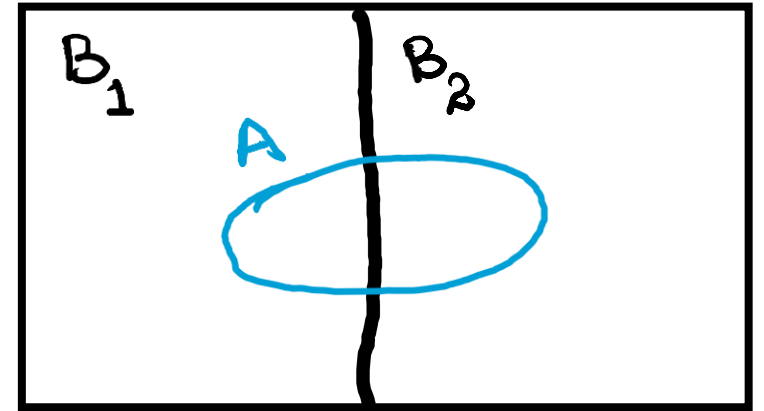
S



1.  $B_1 \cap B_2 = \{\emptyset\}$

# Total probability Law

S

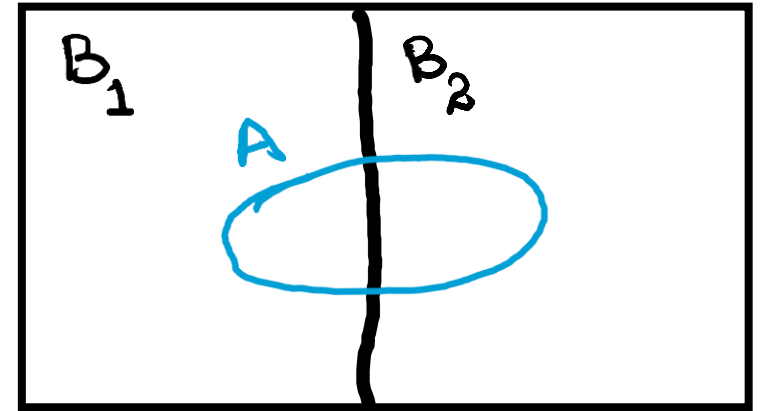


1.  $B_1 \cap B_2 = \{\emptyset\}$

Mutually Exclusive

# Total probability Law

S

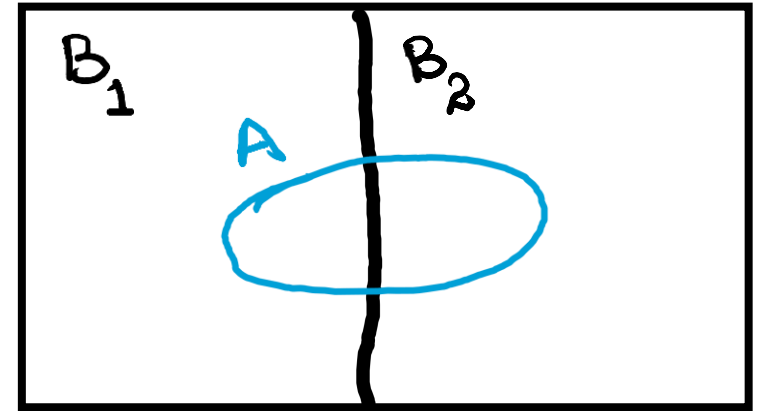


1.  $B_1 \cap B_2 = \{\emptyset\}$   
Mutually Exclusive

2.  $B_1 \cup B_2 = S$

# Total probability Law

S

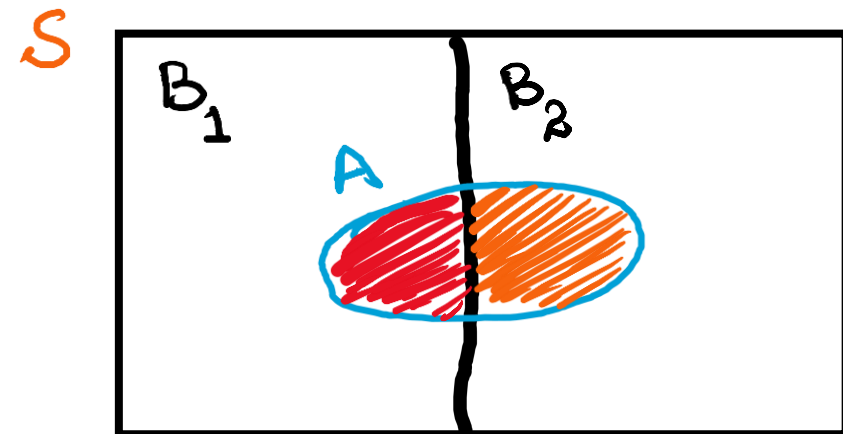


1.  $B_1 \cap B_2 = \{\emptyset\}$   
"Mutually Exclusive"

2.  $B_1 \cup B_2 = S$   
 $\sum_i B_i \rightarrow$  "Partition"

## Total probability Law

$$A = A \cap B_1 + A \cap B_2$$

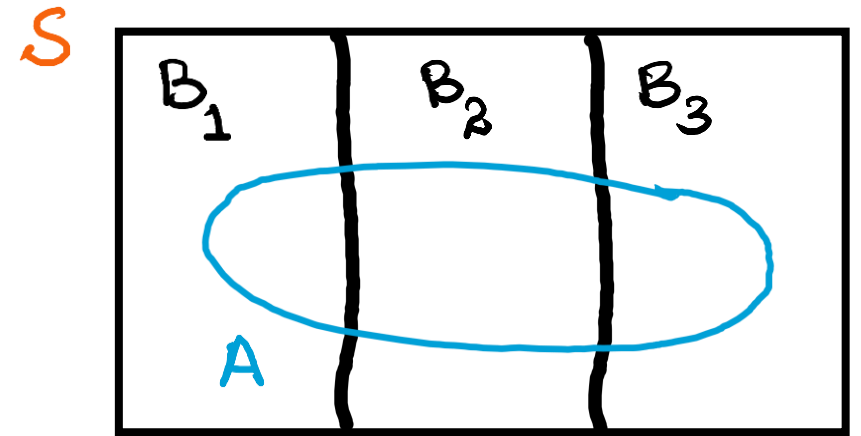


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# Total probability Law

$$A =$$

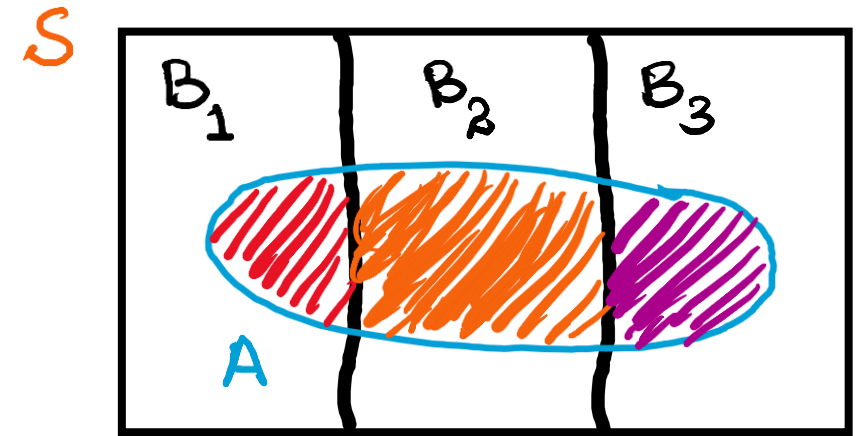


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# Total probability Law

$$A =$$



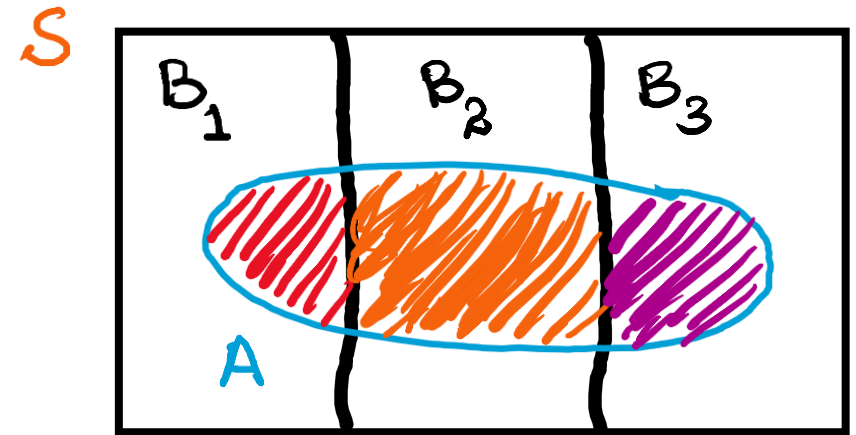
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## Total probability Law

$$A = A \cap B_1$$

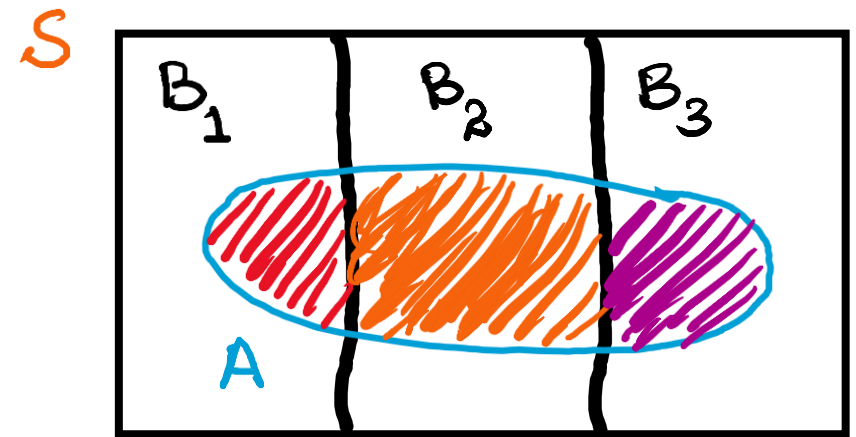


1.  $B_1 \cap B_2 \cap B_3 = \{\emptyset\}$   
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2.  $B_1 \cup B_2 \cup B_3 = S$   
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## Total probability Law

$$A = A \cap B_1 + A \cap B_2$$

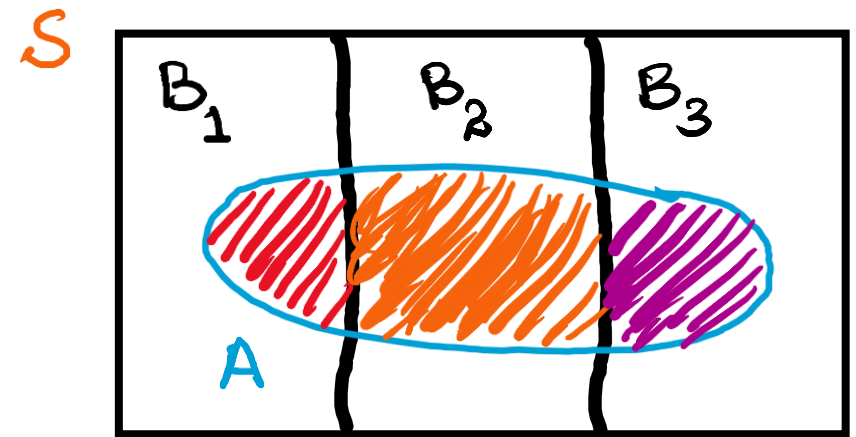


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## Total probability Law

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3$$

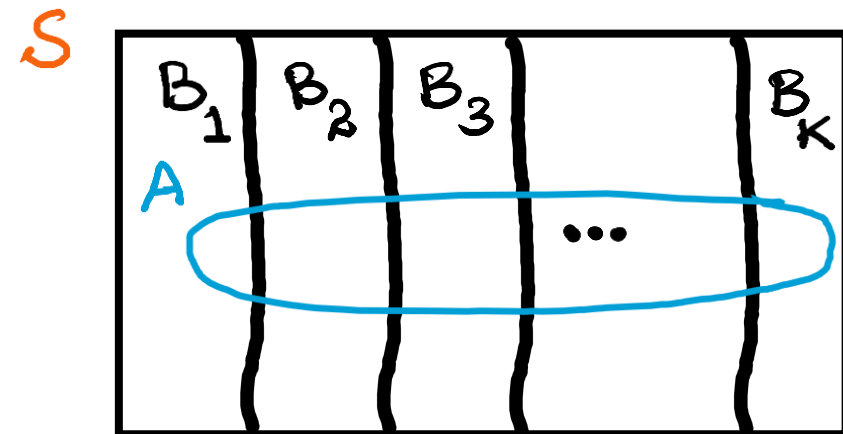


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$$A =$$

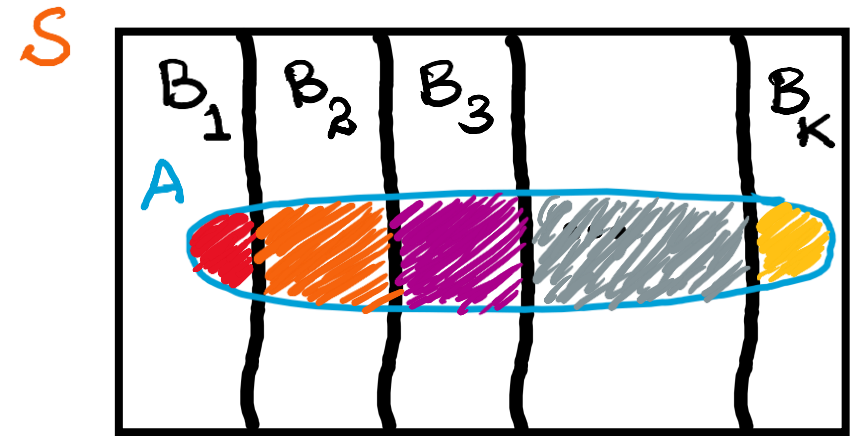


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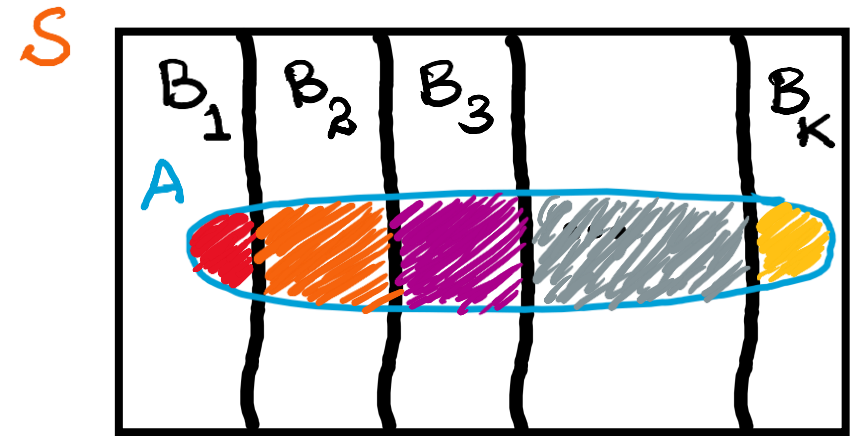


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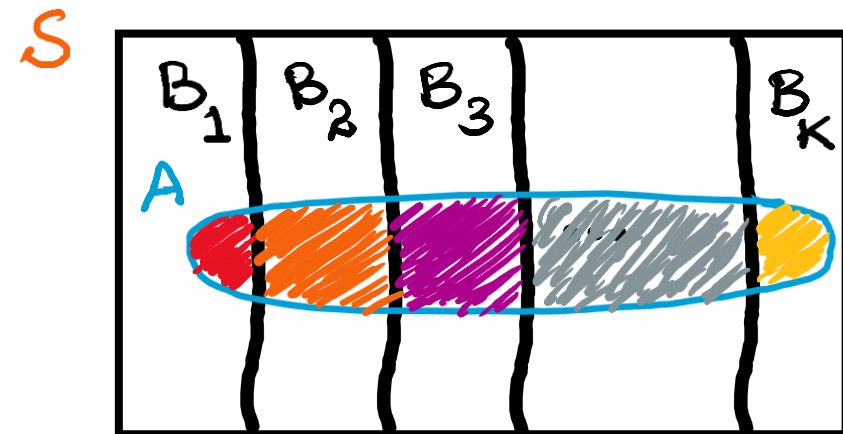


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## Total probability Law

$$A = A \cap B_1 + A \cap B_2$$

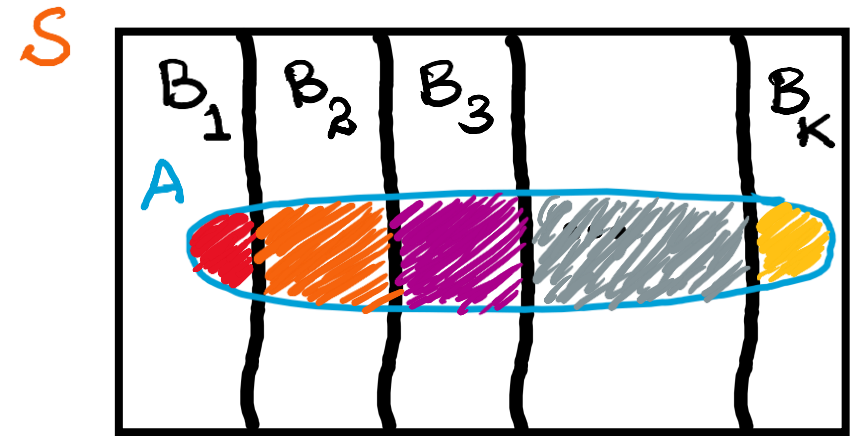


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## Total probability Law

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3$$



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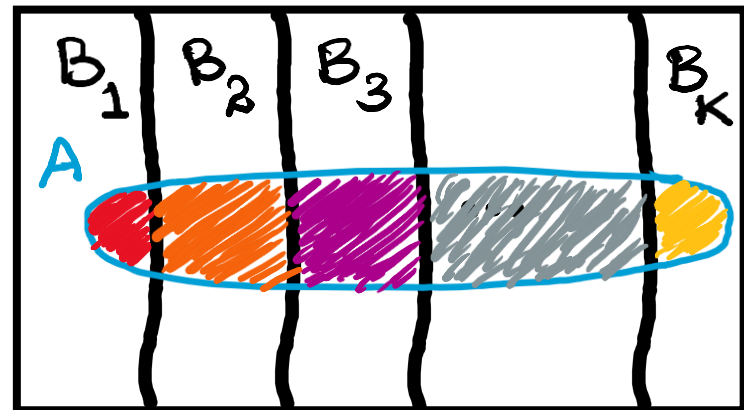
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## Total probability Law

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K$$

S



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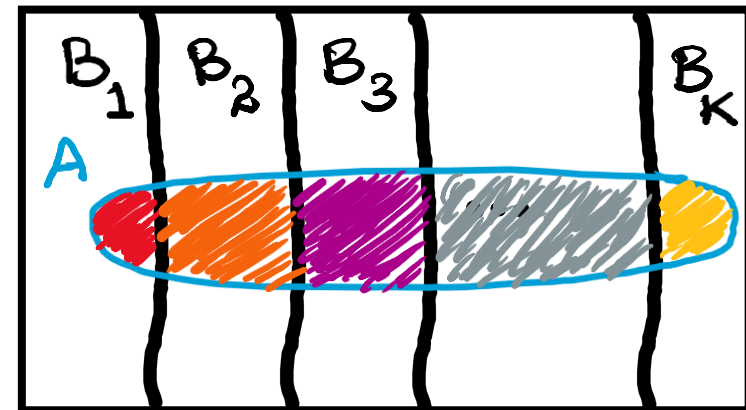
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## Total probability Law

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K$$

$$P(A) =$$

S



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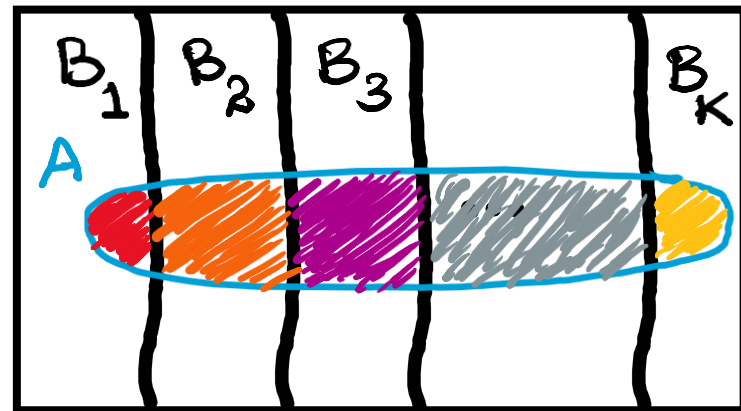
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## Total probability Law

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K$$

$$P(A) = P(A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K)$$

S



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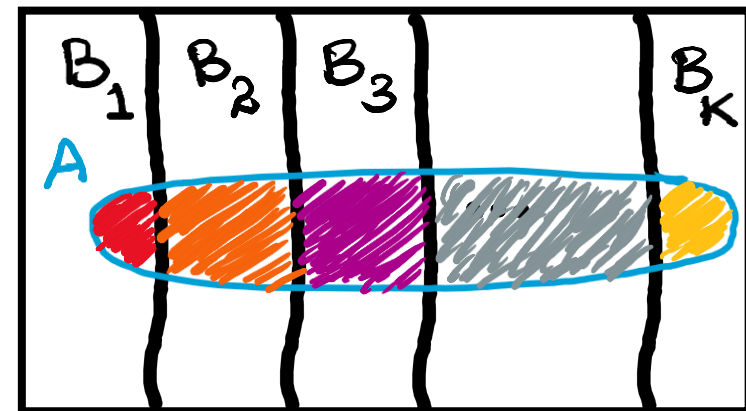
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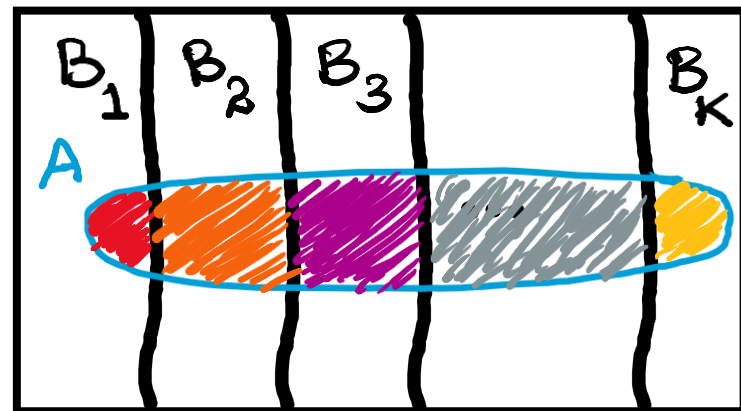
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$$P(A) = P(A \cap B_1)$$

S



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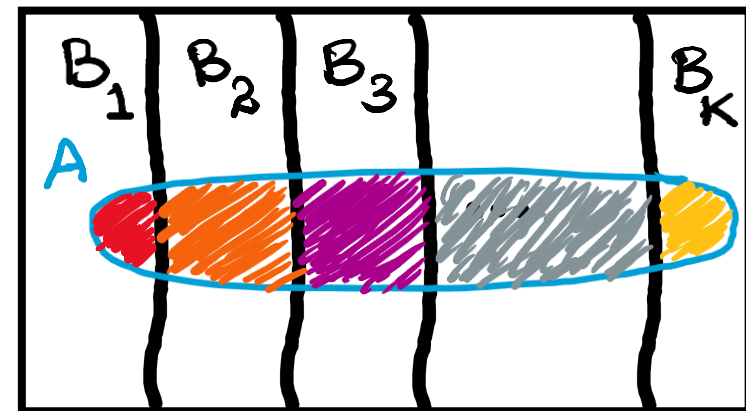
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S



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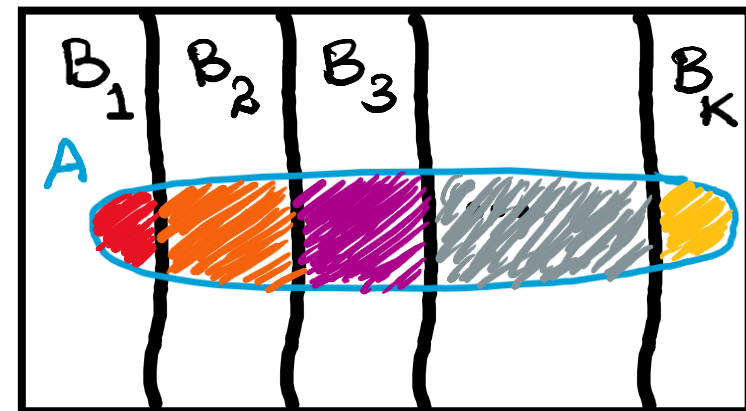
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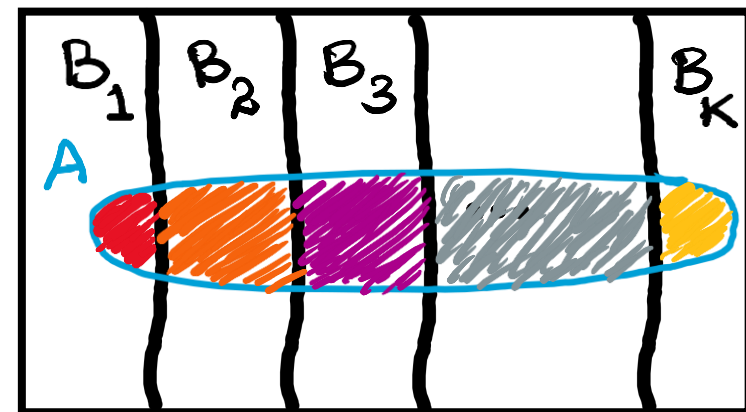
## Total probability Law

S

$$A = A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K$$

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## Total probability Law

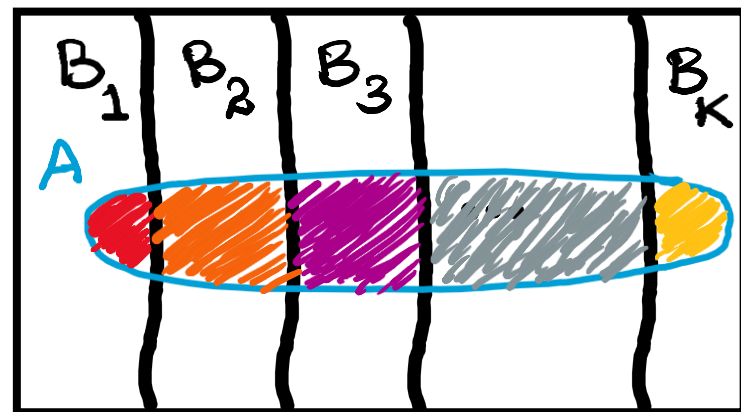
S

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$$P(A) =$$



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## Total probability Law

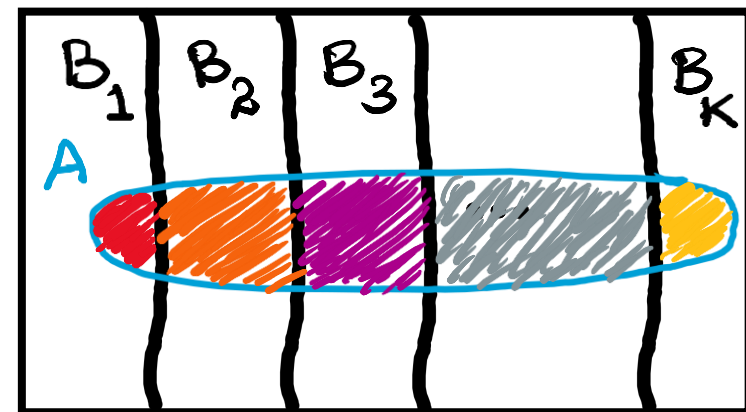
S

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$$P(A) = \sum_{i=1}^K P(A \cap B_i)$$



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 $\sum_i B_i \rightarrow$  "Partition"

## Total probability Law

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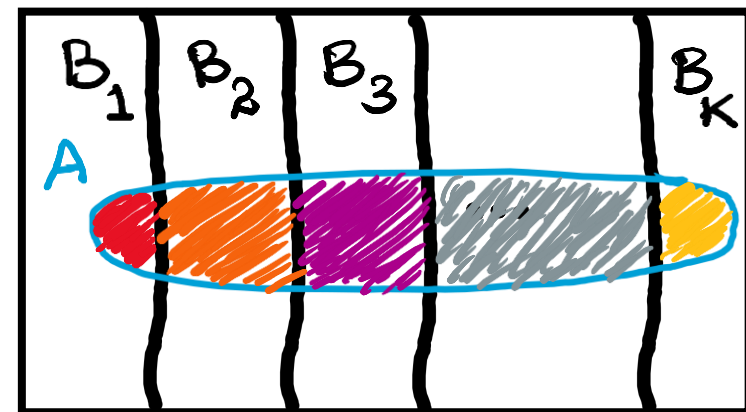
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$$P(A) =$$



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 $\sum_i B_i \rightarrow$  "Partition"

## Total probability Law

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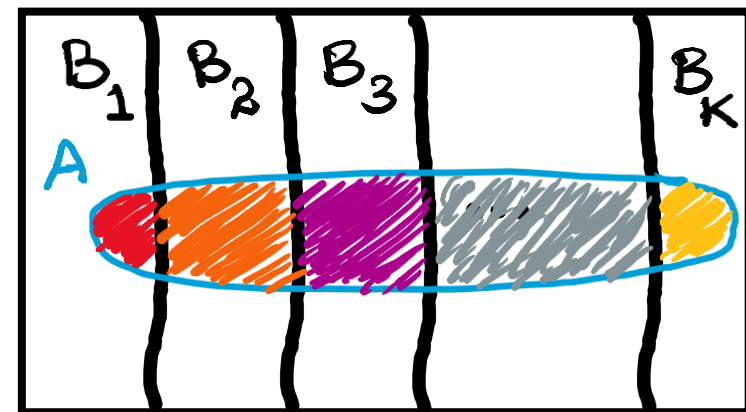
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$$P(A) = \sum_{i=1}^K$$



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"Mutually Exclusive"

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## Total probability Law

S

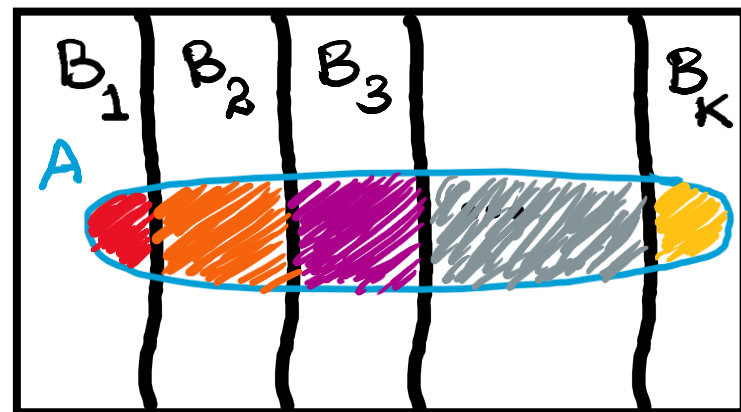
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$$P(A) = P(A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + \dots + P(A \cap B_K)$$

$$P(A) = \sum_{i=1}^K P(A \cap B_i)$$

$$P(A) = \sum_{i=1}^K P(A/B_i)$$



1.  $B_1 \cap B_2 \cap B_3 = \{\emptyset\}$   
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2.  $B_1 \cup B_2 \cup B_3 = S$   
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## Total probability Law

S

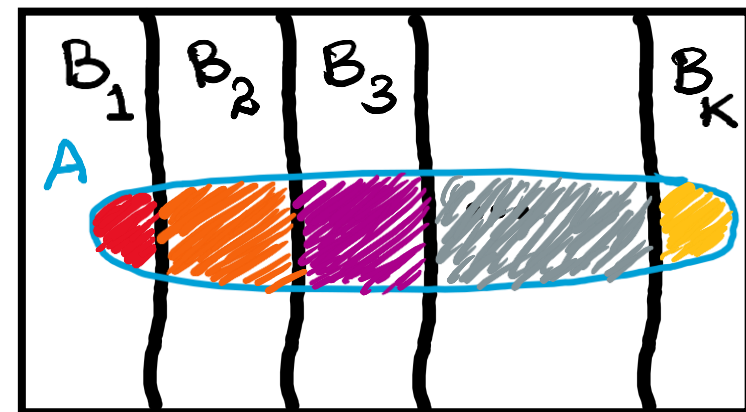
$$A = A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K$$

$$P(A) = P(A \cap B_1 + A \cap B_2 + A \cap B_3 + \dots + A \cap B_K)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + \dots + P(A \cap B_K)$$

$$P(A) = \sum_{i=1}^K P(A \cap B_i)$$

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i) \rightarrow \text{Total probability Law}$$



1.  $B_1 \cap B_2 \cap B_3 = \{\emptyset\}$   
"Mutually Exclusive"

2.  $B_1 \cup B_2 \cup B_3 = S$   
 $\sum_i B_i \rightarrow$  "Partition"



In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?



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*A: product defective*

*B<sub>1</sub>: product by machine B<sub>1</sub>*

*B<sub>2</sub>: product by machine B<sub>2</sub>*

*B<sub>3</sub>: product by machine B<sub>3</sub>*



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$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability that it is defective?**

$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$

$A$ : product defective

$B_1$ : product by machine  $B_1$

$B_2$ : product by machine  $B_2$

$B_3$ : product by machine  $B_3$

$$P(A) = \sum_{i=1}^K P(B_i)P(A/B_i)$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability that it is defective?**

$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$

$$P(A) =$$

*A: product defective*

*B<sub>1</sub>: product by machine B<sub>1</sub>*

*B<sub>2</sub>: product by machine B<sub>2</sub>*

*B<sub>3</sub>: product by machine B<sub>3</sub>*

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



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$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$
$$P(A) = (0.3) + (0.45) + (0.25)$$

*A: product defective*

*B<sub>1</sub>: product by machine B<sub>1</sub>*

*B<sub>2</sub>: product by machine B<sub>2</sub>*

*B<sub>3</sub>: product by machine B<sub>3</sub>*

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$$P(A) = \sum_{i=1}^K P(B_i)P(A/B_i)$$



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$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$
$$P(A) = (0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02)$$

**A:** product defective

**B<sub>1</sub>:** product by machine B<sub>1</sub>

**B<sub>2</sub>:** product by machine B<sub>2</sub>

**B<sub>3</sub>:** product by machine B<sub>3</sub>

$$P(A) = \sum_{i=1}^K P(B_i)P(A/B_i)$$





In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%** and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability that it is defective?**

$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$

$$P(A) = (0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02)$$

$$P(A) = 0.024 \text{ Ans}$$

**A:** product defective

**B<sub>1</sub>:** product by machine B<sub>1</sub>

**B<sub>2</sub>:** product by machine B<sub>2</sub>

**B<sub>3</sub>:** product by machine B<sub>3</sub>

$$P(A) = \sum_{i=1}^K P(B_i)P(A/B_i)$$





A large industrial firm uses three local motels to provide overnight accommodations for its clients. From past experience it is known that 20% of the clients are assigned rooms at the Ramada Inn, 50% at the Sheraton, and 30% at the Lakeview Motor Lodge. If the plumbing is faulty in 5% of the rooms at the Ramada Inn, in 4% of the rooms at the Sheraton, and in 8% of the rooms at the Lakeview Motor Lodge, what is the probability that a client will be assigned a room with faulty plumbing.



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$A$ : faulty plumbing

$B_1$ : Ramad Inn

$B_2$ : Sheraton

$B_3$ : Lakeview motor

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



From past experience it is known that **20% of the clients are assigned rooms at the Ramada Inn**, **50% at the Sheraton**, and **30% at the Lakeview Motor Lodge**. If the **plumbing is faulty in 5% of the rooms at the Ramada Inn**, **in 4% of the rooms at the Sheraton**, and **in 8% of the rooms at the Lakeview Motor Lodge**, **what is the probability that a client will be assigned a room with faulty plumbing**.

$$P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$$

$A$ : faulty plumbing

$B_1$ : Ramad Inn

$B_2$ : Sheraton

$B_3$ : Lakeview motor

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



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$$\begin{aligned} P(A) &= P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3) \\ &\quad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \\ &= (0.02) \qquad \qquad + (0.5) \qquad \qquad + (0.3) \end{aligned}$$

A: faulty plumbing

$B_1$ : Ramad Inn

$B_2$ : Sheraton

$B_3$ : Lakeview motor

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



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$$\begin{aligned} P(A) &= P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3) \\ &\quad \downarrow \\ &= (0.2)(0.05) + (0.5)(0.04) + (0.3)(0.08) \end{aligned}$$

A: faulty plumbing

$B_1$ : Ramad Inn

$B_2$ : Sheraton

$B_3$ : Lakeview motor

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



From past experience it is known that **20% of the clients are assigned rooms at the Ramada Inn**, **50% at the Sheraton**, and **30% at the Lakeview Motor Lodge**. If the **plumbing is faulty in 5% of the rooms at the Ramada Inn**, in **4% of the rooms at the Sheraton**, and in **8% of the rooms at the Lakeview Motor Lodge**, **what is the probability that a client will be assigned a room with faulty plumbing.**

$$\begin{aligned} P(A) &= P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3) \\ &= (0.2)(0.05) + (0.5)(0.04) + (0.3)(0.08) \\ &= 0.054 \text{ Ans} \end{aligned}$$

A: faulty plumbing

$B_1$ : Ramad Inn

$B_2$ : Sheraton

$B_3$ : Lakeview motor

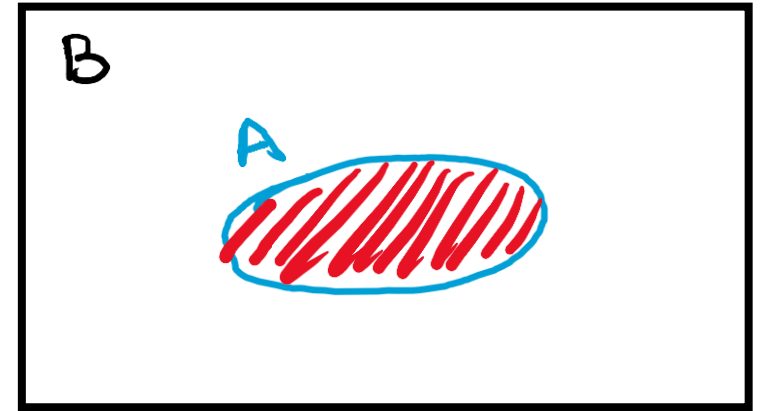
$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$

# Bayes' Rule



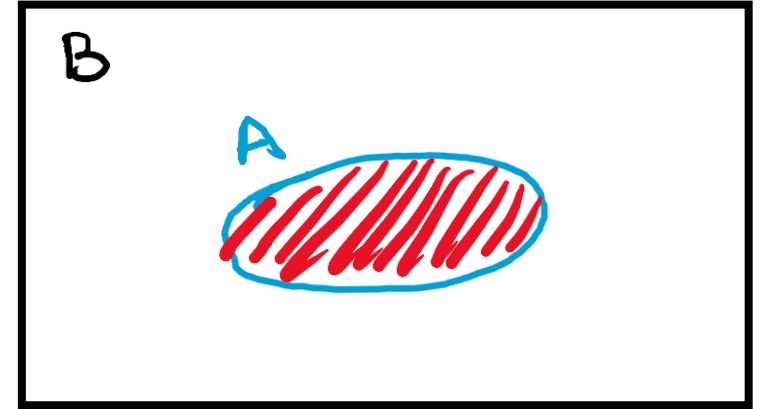
## Bayes' Rule

$$B \cap A = A \cap B$$



## Bayes' Rule

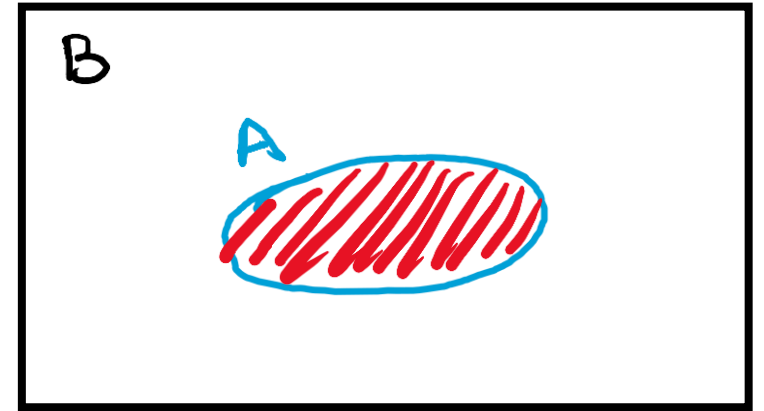
$$P(B \cap A) = P(A \cap B)$$



## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

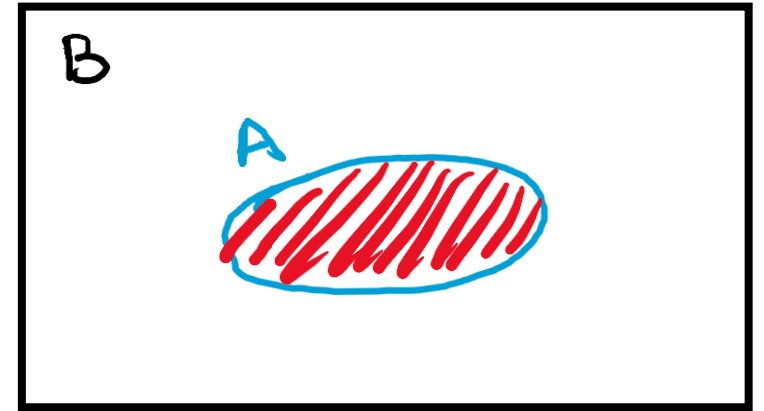
$$P(B/A)$$



## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

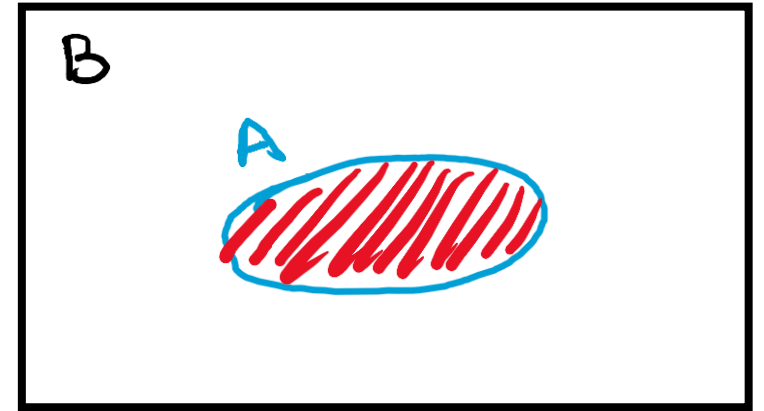
$$P(A) P(B/A) =$$



## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

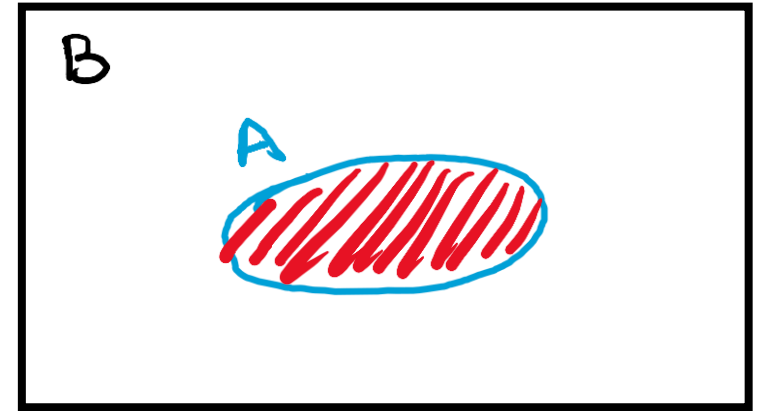
$$P(A) P(B/A) = P(A/B)$$



## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

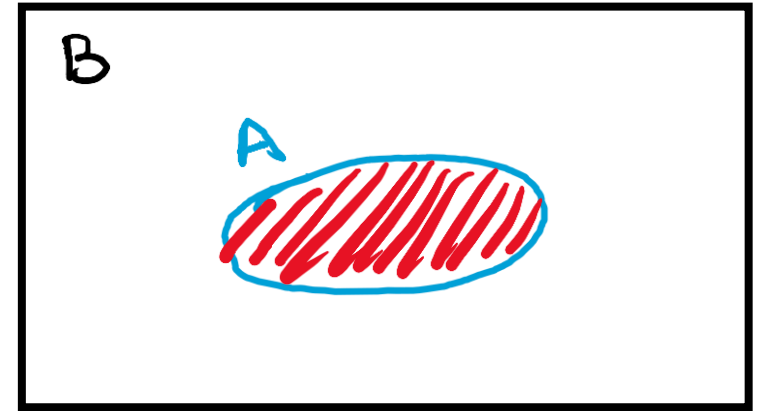


## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$



## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

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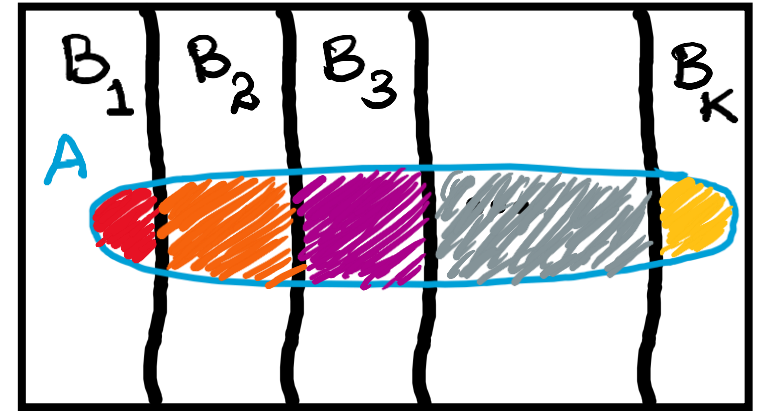
$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$

## Bayes' Rule

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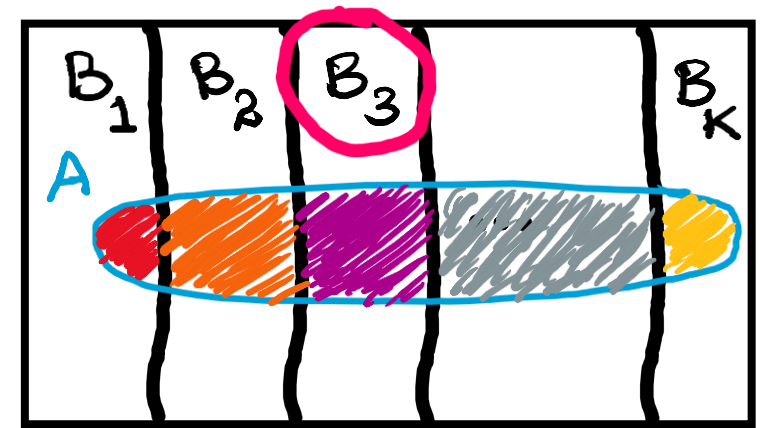


## Bayes' Rule

$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

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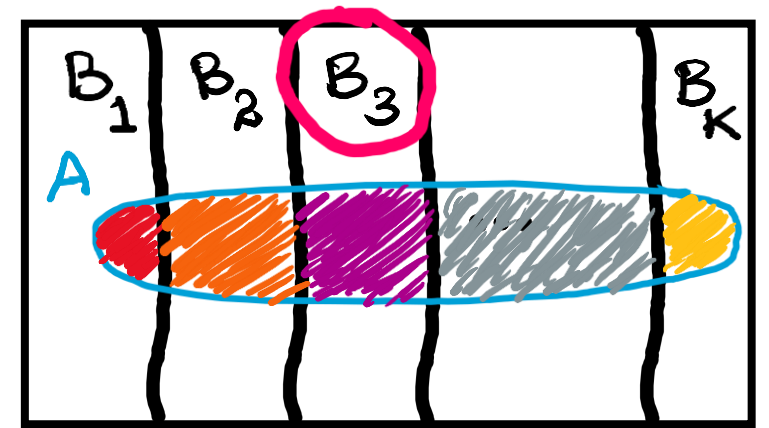


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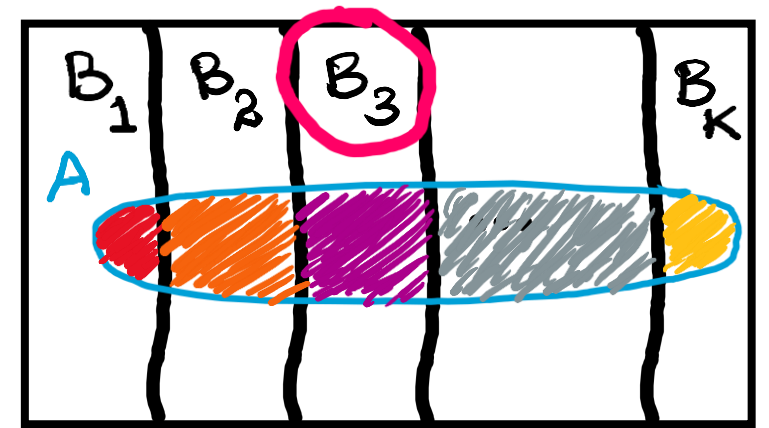
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## Bayes' Rule

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$$P(A) P(B/A) = P(B) P(A/B)$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$



$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

## Bayes' Rule

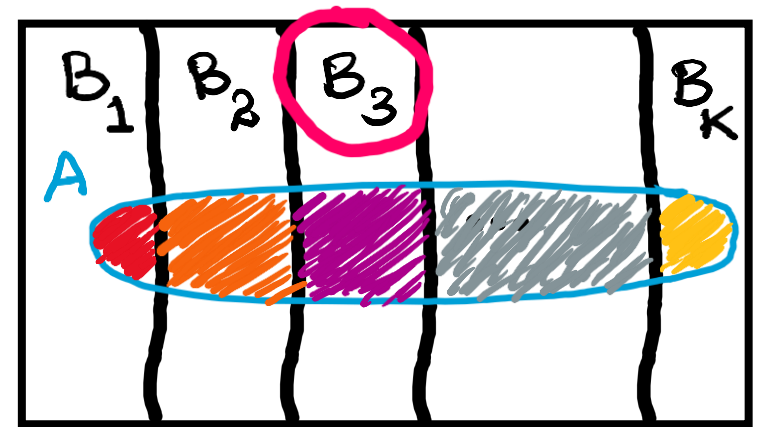
$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$

Total Probability Law

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

## Bayes' Rule

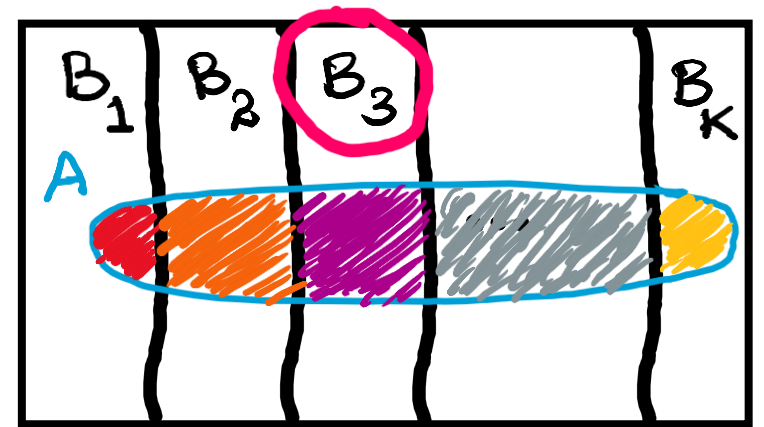
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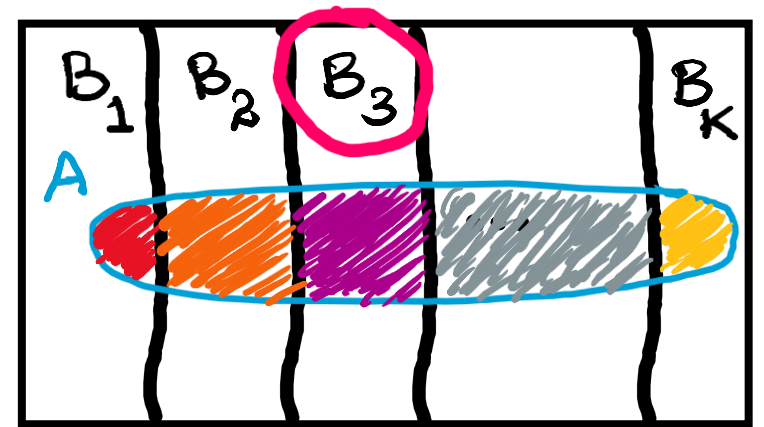
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$$P(A) P(B/A) = P(B) P(A/B)$$

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Total Probability Law

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{\sum_{i=1}^K P(B_i) P(A/B_i)}$$



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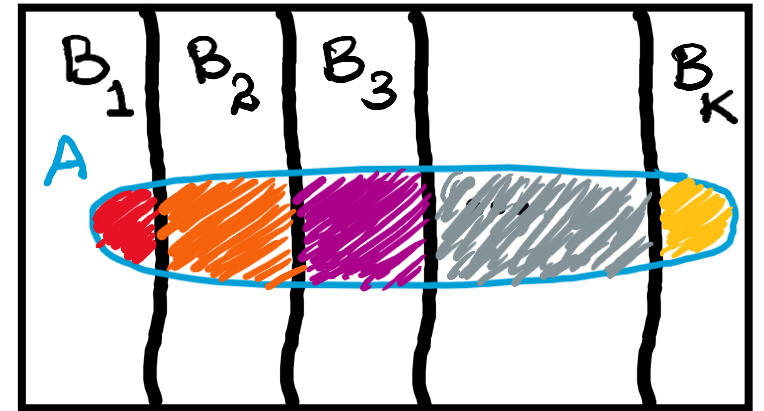
$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

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Total Probability Law

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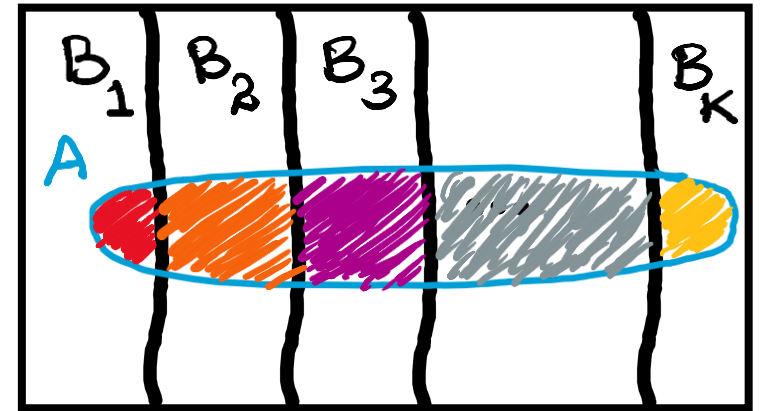
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Total Probability Law

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$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

$$P(B_i/A) = \frac{P(B_i) P(A/B_i)}{\sum_{i=1}^K P(B_i) P(A/B_i)}$$

## Bayes' Rule

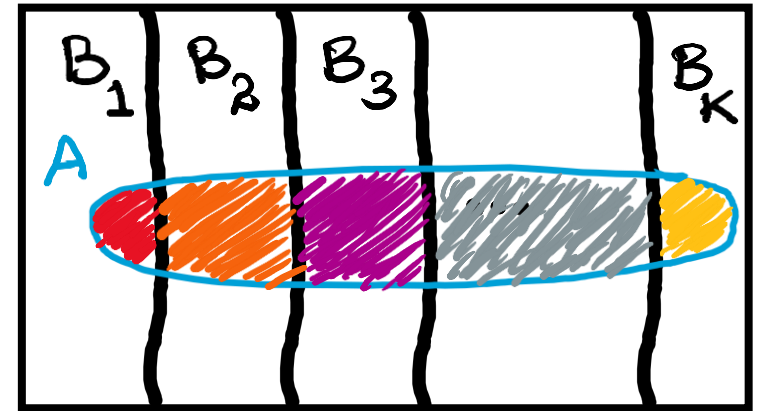
$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$

Total Probability law

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

$$P(B_\gamma/A) = \frac{P(B_\gamma) P(A/B_\gamma)}{\sum_{i=1}^K P(B_i) P(A/B_i)}$$

$$\gamma = 1, 2, \dots, K$$

## Bayes' Rule

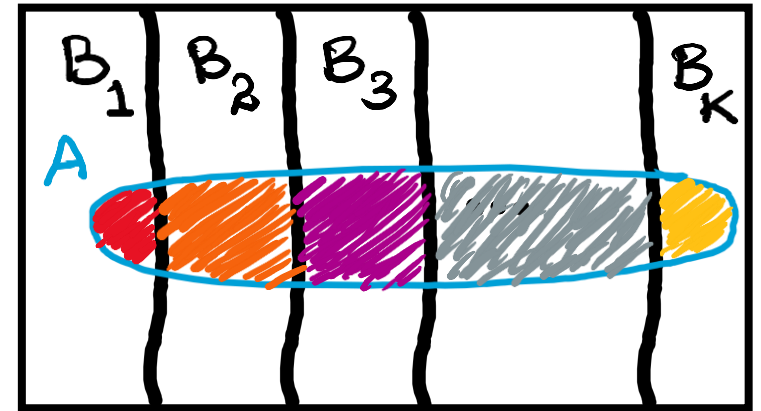
$$P(B \cap A) = P(A \cap B)$$

$$P(A) P(B/A) = P(B) P(A/B)$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$

Total Probability Law

$$P(A) = \sum_{i=1}^K P(B_i) P(A/B_i)$$



$$P(B_3/A) = \frac{P(B_3) P(A/B_3)}{P(A)}$$

$$P(B_i/A) = \frac{P(B_i) P(A/B_i)}{\sum_{i=1}^K P(B_i) P(A/B_i)}$$

$i = 1, 2, \dots, K$

↓  
Bayes' Rule



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**



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$A$ : product defective

$B_1$ : product by machine  $B_1$

$B_2$ : product by machine  $B_2$

$B_3$ : product by machine  $B_3$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

$A$ : product defective

$B_1$ : product by machine  $B_1$

$B_2$ : product by machine  $B_2$

$B_3$ : product by machine  $B_3$

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

$$P(B_3|A) =$$

$A$ : product defective

$B_1$ : product by machine  $B_1$

$B_2$ : product by machine  $B_2$

$B_3$ : product by machine  $B_3$

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$





In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

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$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1)}$$

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)}$$

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)}$$

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$



In a certain assembly plant, three machines, **B1**, **B2**, and **B3**, make **30%**, **45%**, and **25%**, respectively, of the products. It is known from past experience that **2%**, **3%**, and **2%** of the **products made by each machine, respectively, are defective**. Now, suppose that a finished product is randomly selected. **What is the probability it was made by machine B3?**

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=

$$P(B_1) = 0.3$$

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$$\begin{aligned} P(B_1) &= 0.3 & P(A|B_1) &= 0.02 \\ P(B_2) &= 0.45 & P(A|B_2) &= 0.03 \\ P(B_3) &= 0.25 & P(A|B_3) &= 0.02 \end{aligned}$$

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$$= \frac{(0.25)(0.02)}{(0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02)}$$

=

$$P(B_1) = 0.3 \quad P(A|B_1) = 0.02$$

$$P(B_2) = 0.45 \quad P(A|B_2) = 0.03$$

$$P(B_3) = 0.25 \quad P(A|B_3) = 0.02$$

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$$= \frac{(0.25)(0.02)}{(0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02)}$$

$$= \frac{10}{49} \text{ Ans}$$

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A certain federal agency employs three consulting firms (A, B, and C) with probabilities 0.40, 0.35, and 0.25, respectively. From past experience it is known that the probability of cost overruns for the firms are 0.05, 0.03, and 0.15, respectively. Suppose a cost overrun is experienced by the agency. What is the probability that the consulting firm involved is company C?



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O: Overrun



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O : Overrun

A : consulting firm A



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**O** : Overrun

**A** : consulting firm A

**B** : consulting firm B



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A : consulting firm A

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O: Overrun

$P(A) = 0.4 \leftarrow A : \text{consulting firm A}$

$P(B) = 0.35 \leftarrow B : \text{consulting firm B}$

$P(C) = 0.25 \leftarrow C : \text{consulting firm C}$



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$$P(O/A) = 0.05$$

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$$P(O/C) = 0.15$$

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O: Overrun

$$P(C/O) = ?$$

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$$P(O/B) = 0.03$$

$$P(O/C) = 0.15$$

$$P(A) = 0.4 \leftarrow A : \text{consulting firm A}$$

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Bayes' Rule

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{\sum_{i=1}^K P(B_i)P(A|B_i)}$$



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Bayes' Rule

$$P(C/O) = \frac{P(C) P(O/C)}{P(O)}$$

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$$P(C/O) = \frac{P(C)P(O/C)}{P(A)P(O/A) + P(B)P(O/B) + P(C)P(O/C)} = 0.55 \text{ Ans}$$

A manufacturing firm employs three analytical plans for the design and development of a particular product. For cost reasons, all three are used at varying times. In fact, plans 1, 2, and 3 are used for 30%, 20%, and 50% of the products, respectively. The defect rate is different for the three procedures as follows:

$$P(D|P_1) = 0.01, \quad P(D|P_2) = 0.03, \quad P(D|P_3) = 0.02,$$

where  $P(D|P_j)$  is the probability of a defective product, given plan  $j$ . If a random product was observed and found to be defective, which plan was most likely used and thus responsible?